



HOLOTRADE

An Exchange for Compute

Node-second settlement with explicit energy, workload-state, health, capability, locality, and basket adjustments

Many compute products still quote a rectangle of capacity for a span of time. Sub-second microVM startup makes finer metering useful, but not costless. Once cold-start cost is itemised, energy interval, workload state, health, capability, locality and basket topology can each be disclosed rather than flattened into one rate.

This paper is about what becomes priceable.

Wil Dahn

Built on the $W(3,3)$ - E_8 substrate programme

github.com/wilcompute/Holotrade

August 2026

How to read the evidence tags. **PROVED** means an exact finite claim with a repository witness or exhaustive check. **MEASURED** means an observed property of the implemented prototype or a tool output such as synthesis; it does not mean production-market or physical-hardware evidence. **MODELLED** means a simulation output, policy coefficient or estimate. **PUBLISHED** means an external result and is accompanied by a citation. A passing regression test establishes that code follows its specification; it is not, by itself, evidence that an economic or physical model describes the world.

Abstract

Holotrade is a research prototype for quoting and settling compute at node-second granularity while exposing energy, workload-state, utilisation, health, capability and locality adjustments separately. It combines three artifacts: a six-factor quote engine with an explicit cost floor, a seeded fleet simulation, and a table-free adjacency primitive based on the 40-point collinearity graph of $W(3,3)$. The finite graph facts and Boolean circuit equivalences are exact within their stated encodings. The fleet, prices, “genomes”, wear, order flow and recursive fabric are models, not observations from a production venue. In a paired 64-seed experiment with 220 simulated nodes, the two-sided demand rule reduced terminal utilisation dispersion in all 64 runs; this establishes behaviour of the model, not longer hardware life or market efficacy. No physical photonic substrate or live exchange exists.

Contributions and evidence taxonomy

Tag	Supports	Does not support
PROVED	Exhaustive finite checks, explicit witnesses and SAT equivalence for the encoded circuits	Physical performance, economic adoption or an unstated generalisation beyond the checked object
MEASURED	Prototype behaviour and reproducible tool output	Production telemetry, customer outcomes or placed-and-routed hardware
MODELLED	Seeded simulations, policy rules, coefficients and conditional estimates	Empirical causal conclusions
PUBLISHED	A result in the cited primary source	Claims outside that source’s assumptions or measurement environment

The concrete contributions are: (i) an auditable quote decomposition and node-second settlement prototype; (ii) an exact implementation of the $W(3,3)$ adjacency graph and a separately scoped recursive address-distance model; and (iii) a software/RTL admission-policy artifact with explicit refusal reasons. The sections below keep those contributions separate from proposed market, security and hardware deployments.

Contents

Abstract	2
Contributions and evidence taxonomy	2
1 The thesis, in one page	5
2 What is actually being sold	7
2.1 The case against the hour	7
2.2 Three objects, not one	8
2.3 Why isolation is a precondition for the market existing at all	8
3 The price, term by term	9
3.1 E — energy	9
3.2 G — genetics	9
3.3 D — demand and wear: the balancer	10
3.4 H — health	12
3.5 Q — quantum	12
3.6 L — locality	12
3.7 The floor: what the exchange refuses to do	13
4 The network is the computer	14
4.1 The identity	14
4.2 The consequence: you buy shapes, not counts	15
4.3 Which topologies are actually optimal, and can they be delivered	15
4.3.1 Delivery: guarantees rather than averages	17
4.3.2 Fragmentation: the sizes that leave nothing stranded	17
4.4 What happens above one cell, and why it does not scale	18
4.4.1 Every optimal shape above level one is a replication	18
4.5 The mechanism this creates: a defragmentation market	19
4.6 The recursive address model	20
4.7 Joining is local in the prototype	20
5 Instruments	22
5.1 A policy mobility score, not observed liquidity	23

6	Energy and the thermodynamic floor	24
6.1	The honest efficiency denominator	24
7	The hardware	25
7.1	It is proved, not just written	25
7.2	Current synthesis footprint	25
7.3	The gate refuses rather than degrades	26
8	What exists today	27
8.1	The reproducibility contract	27
8.2	Two bugs the tests caught, and why that is the point	28
9	Honest scope	29
9.1	The risks worth stating	29
A	Glossary	31
B	Selected constants and parameters	32
C	References	32

1 The thesis, in one page

Compute is increasingly traded as standardized capacity, while many products still display a fixed hourly-equivalent rate for a broadly specified machine. That convention suppresses information in four directions at once.

1. **It destroys the energy signal.** Wholesale electricity prices move every five minutes in ERCOT's real-time market and can be negative [8]. An hourly quote cannot expose that interval-level input. A node-second meter can associate each second with the grid-price interval that contains it; it does not create a new one-second wholesale price.
2. **It destroys the machine's history.** Two identical servers that have run different work may carry different caches, checkpoints, tuned kernels or scheduling state. Whether that state improves a future job is an empirical question. Holotrader represents the hypothesis explicitly so it can be tested rather than assuming that the physical chassis itself has learned.
3. **It destroys the wear signal.** A heavily used machine may be accumulating temperature excursions and usage-dependent wear, while a cold machine still draws standby power and ages on the calendar. A single rate does not expose either signal; the prototype models them separately, without claiming a validated lifetime effect.
4. **It destroys the shape.** Forty machines scattered across a datacentre and forty machines connected as one low-diameter block can have the same count but different usable connectivity. Holotrader therefore computes a separate basket-level topology score rather than treating it as a node property.

Holotrader's node quote exposes all four, together with capability and locality. The prototype displays an hourly-equivalent rate so buyers can compare familiar prices, then prorates that rate to the second at settlement:

$$P^{\text{hr}} = P_0 \times E \times G \times D \times H \times Q \times L$$

hourly-equivalent quote; settlement rate is $P^{\text{hr}}/3,600$ per node-second

The unit that settles is the **node-second**. The current model uses a 171.5 ms cold-start baseline, informed by the Firecracker microVM architecture [1]. **MODELLED** A fresh boot therefore consumes 17.15% of a one-second allocation and leaves 82.85% for useful work; boot overhead falls below 1% only once useful runtime exceeds roughly 17 seconds. Holotrader charges that cold start once and separately rather than concealing it in an hourly rounding rule.

The paired simulation result motivating the balancer

The demand term D is *two-sided*: it charges a premium above a target utilisation band and offers a discount below it. Across 64 paired seeds, with 220 simulated nodes and 500 one-minute steps (8.333 simulated hours per arm), mean terminal utilisation Gini was **0.0645846** with the balancer on (95% CI [0.0604334, 0.0687358]) and **0.1610963** with it off (95% CI [0.1545541, 0.1676386]). Mean paired difference was **0.0965118** (95% CI [0.0930282,

0.0999953]); mean paired relative reduction was **0.6033545** (95% CI [0.5889816, 0.6177275]). All 64 paired runs improved. **MODELLED**

This is a reproducible property of the stated demand-response model. It is not a measurement of a live market, temperature cycling, failure rate or hardware lifetime. Those consequences require telemetry and a separately validated wear model.

Everything else in this paper follows from taking those two moves seriously: a sub-second unit, and a price with observable parts.

2 What is actually being sold

2.1 The case against the hour

Many incumbent products quote node-hours. The hour is a commercial display and reservation convention, not a unit of computation. Faster provisioning makes finer settlement useful, but it does not eliminate metering, reconciliation, minimum-charge or cold-start costs.

IN PLAIN ENGLISH: What a virtual machine is

A physical server is one piece of hardware. A *virtual machine* is a software-created illusion of a whole separate computer running on top of it, so that one physical server can be rented to several customers at once without them being able to see or interfere with each other. The software that creates the illusion is called a *hypervisor*. Creating a traditional virtual machine involves booting a complete operating system inside the illusion, which is why it takes minutes.

IN PLAIN ENGLISH: What a microVM is, and why it changes the economics

A *microVM* is a stripped-down virtual machine: a real hypervisor boundary, a real separate operating-system kernel, but with everything unnecessary for one specific job removed — no virtual graphics card, no virtual sound card, no legacy device emulation. It gets the same security isolation as a full virtual machine with a reduced device model. Amazon's *Firecracker* is the best-known implementation and underlies AWS serverless products [1].

The prototype uses **171.5 ms** as its cold-start baseline. **MODELLED** Firecracker's published design and performance evaluation establish the relevance of sub-second microVM startup, but this paper does not present the 171.5 ms value as a new measurement or as portable across storage, kernel and host configurations [1]. Three design consequences follow.

1. **The hour has no physical justification.** A one-second billing slice carries 17.15% startup overhead in the worst case where every second is a fresh boot. The fraction is below 1% only beyond roughly 17 seconds of useful runtime. This supports itemised cold-start charging and second-level metering; it does not make startup free.
2. **Per-second energy pricing becomes possible rather than visible.** ERCOT publishes real-time prices on five-minute intervals [8]. Holotrader meters execution by the second and associates each metered second with its containing price interval; it does not interpolate a one-second wholesale price.
3. **The startup cost becomes visible instead of smeared.** Holotrader charges the boot *once, explicitly, as its own line*, rather than folding it into the rate. A customer running many short jobs can therefore see exactly what their fan-out pattern costs them — which is information the hourly model structurally cannot provide.

2.2 Three objects, not one

Layer	Object	What it is
ASSET	the node	A durable machine with an identity, a modeled workload-state record, and a health record. This is what you own, lease, or lend into the pool.
CONTRACT	the execution plan	A document specifying intended software artefacts, permissions and a validity window. The prototype demonstrates an integrity seal and admission predicates; production digital signatures and attestation are future work.
UNIT	the node-second	What actually settles. Each second is associated with the wholesale-price interval in force when it was drawn.

IN PLAIN ENGLISH: What an execution plan is, and why it matters commercially

Think of it as a specified work order. Before any machine starts, the plan states: here is the intended software digest, here are the network grants, and here is the validity window. A production version would sign that canonical content using a standard digital-signature scheme such as those specified by FIPS 186-5 [7], persist replay state, and bind admission to remote attestation. The present browser prototype does not implement those controls.

Commercially this separates a capacity reservation from a specification of intended execution. A signature authenticates that specification; it does not by itself prove what ran. Transferability would additionally require assignment rules, attestation, counterparty controls and jurisdiction-specific legal terms.

2.3 Why isolation is a precondition for the market existing at all

A two-sided marketplace requires that an owner of idle machines be willing to run a stranger's code on them. Without hardware-enforced isolation, "list your spare capacity" is a request to be compromised, and the supply side never materialises.

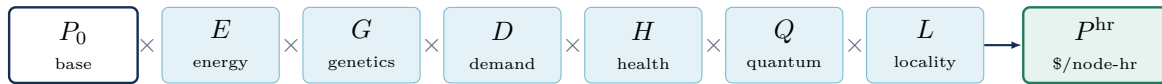
The deployment model Holotrader proposes gives each job its own kernel under a microVM hypervisor, deny-by-default network grants, verified filesystem images, remote attestation and an externally anchored audit log. Firecracker supports the reduced-device microVM boundary [1]; the remaining controls are requirements, not features established by this prototype.

IN PLAIN ENGLISH: What a hash chain is

Each entry in a correctly implemented hash chain commits to its content and the previous entry. Verification must recompute every digest, and the chain head must be retained or externally anchored; otherwise an attacker can rewrite the suffix or the whole chain. The current demo checks continuity and deletion of a middle entry only. It is an integrity-seal demonstration, not a production tamper-proof log.

3 The price, term by term

The formula is multiplicative rather than additive. That keeps each adjustment on its own displayed line and permits separate bounds. It does *not* make the inputs statistically or causally independent: utilisation, workload state, health and energy can be correlated. An explicit floor is applied after the multiplication, and a failed capability predicate makes the quote inadmissible rather than infinite.



each term shown separately; hourly equivalent settles pro rata by the second

3.1 E — energy

IN PLAIN ENGLISH

Datacentres buy electricity on markets whose real-time settlement prices change by interval. ERCOT's real-time market uses five-minute settlement intervals and can publish negative or scarcity prices [8]. E maps the current interval's price into a bounded quote adjustment.

E is deliberately *not* linear in the grid price. A node should not become worthless when the grid goes negative, nor unsellable when it spikes. Holotrader currently applies a power law with exponent 0.55 to a positive-clamped ratio against the site's baseline, then clamps the multiplier to [0.62, 2.40]. **MODELLED** These are policy parameters, not estimated operator behaviour. Negative prices map to the lower clamp. PUE enters the separately displayed energy-cost floor, not this multiplier.

IN PLAIN ENGLISH: What PUE is

Power Usage Effectiveness. For every watt that reaches a computer chip, a datacentre spends additional watts on cooling, power conversion and lighting. PUE is the ratio of total facility power to computing power. A PUE of 1.0 would mean no facility overhead. In this prototype it multiplies the modeled energy cost used in the quote floor.

3.2 G — genetics

This is the term that makes Holotrader a different product rather than a better price list.

IN PLAIN ENGLISH

When a machine runs the same class of work repeatedly, the software layer on it accumulates useful statistical structure — cached compilations, tuned kernel selections, warm model checkpoints, learned scheduling parameters. The industry already exploits this deliberately: “warm starts” and “checkpoint merging” are standard practice in machine-learning operations.

The testable hypothesis is that two physically identical servers carrying different reusable software state may deliver different time-to-result on a specified future workload. Holotrader

calls that versioned, copyable software state the node's *genome*; the name does not imply that the chassis itself learns or that improvement is guaranteed.

G is a model score built from three logged quantities. Their predictive value must be validated out of sample before G can support a production premium:

- **Specialisation** on *this specific* workload class — not a general capability score. A node with excellent memory-bandwidth characteristics gets no credit for it on a workload that is compute-bound.
- **Realised fitness**: the completion record. Jobs finished versus jobs failed, weighted by experience depth. Never a manufacturer's specification.
- **Provenance depth**: lineage depth, treated here as a prior rather than proof of quality.

The caveat that goes on every lease quote

The simulation evolves a leased core toward the workload class it receives. **MODELLED** A production lease would need to define ownership, customer-data isolation, permitted updates, rollback and an out-of-sample benchmark. The quote therefore displays projected model drift as a scenario, not a measured 90-day forecast.

3.3 D — demand and wear: the balancer

IN PLAIN ENGLISH

The obvious way to price demand is one-sided: a busy machine costs more. That is what surge pricing does, and it leaves the cold half of your fleet earning nothing while still aging, still drawing standby power, and still due for the same maintenance visit.

Holotrader charges a premium above a target band *and offers a discount below it*. In the prototype this is a demand-allocation policy. Whether its discount is cheaper than alternative operations is an empirical question.

What the utilisation result does and does not say

Temperature excursions can contribute to package fatigue, while sustained voltage, temperature and current contribute other wear mechanisms. The present simulation tracks utilisation and a modeled health trajectory; it does not measure junction temperature or count thermal cycles.

Lower cross-sectional utilisation Gini means work is distributed more evenly at the observation time. It does not imply lower temporal cycling, lower energy, longer life or de-clustered service events. Those are hypotheses for a telemetry study using temperature traces, failures and maintenance records.

IN PLAIN ENGLISH: What a Gini coefficient is

A single number between 0 and 1 measuring how unevenly something is distributed. Economists use it for income inequality. Here it measures how unevenly work is distributed across the fleet: 0 would mean every machine carries exactly the same load, 1 would mean one machine does everything while the rest sit idle. Lower is better.

Paired 64-seed simulation

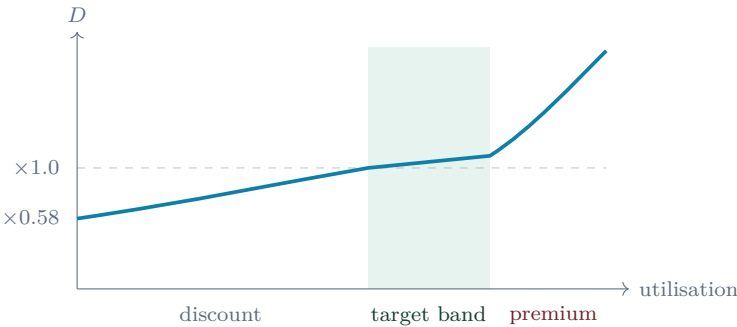
	Mean	95% CI	Paired outcome
Balancer off	0.1610963	[0.1545541, 0.1676386]	
Balancer on	0.0645846	[0.0604334, 0.0687358]	64/64 lower
Paired difference	0.0965118	[0.0930282, 0.0999953]	off minus on
Relative reduction	0.6033545	[0.5889816, 0.6177275]	paired by seed

Each pair uses the same seed, 220 nodes and 500 one-minute steps, changing only whether *D* is enabled; each arm spans 8.333 simulated hours. The aggregate target level is anchored to the centre of the policy band; relative prices within each hardware class determine the cross-sectional allocation. The confidence intervals are across the 64 seeded runs. The frozen summary is `data/balancer_ab_64.json`, generated by `node experiments/balancer_ab.js -summary`. MODELLED

Two engineering details matter for whether this is honest.

The comparison set is the hardware class, not the fleet. A buyer choosing between accelerators does not consider a general-purpose CPU node “cheap” — they are not substitutes. Comparing against a fleet-wide median would make every CPU look like a permanent bargain and every scarce node look permanently extortionate, and the loop would never settle.

The curve must be monotone. If there is any utilisation at which a node gets *cheaper* by becoming busier, a buyer can arbitrage the discount by loading the machine they are about to rent. An earlier version of this design had exactly that defect at one of the seams between the discount, band and premium regions. It was caught by a monotonicity test, and the test is in the repository.



The premium region is superlinear by policy. Its current wear surcharge uses utilisation, an exponential moving average and a hardware sensitivity coefficient. MODELLED It is not a calibrated

damage law; production pricing would require temperature and reliability evidence before describing the premium as a maintenance reserve caused by a specific buyer.

3.4 H — health

H combines a modeled performance derate, Weibull hazard and correctable-error score. **MODELLED** The terms are displayed separately in the implementation, but their coefficients are not calibrated against production failures and can overlap with maintenance reserve and capital recovery.

IN PLAIN ENGLISH: What a correctable error is, and why it predicts failure

ECC memory can detect and correct specified error patterns. Large-scale field data show that correctable errors are associated with elevated future error risk, but not that every rising count predicts imminent machine failure [4]. Holotrader treats the count as a candidate feature; a production model must control for DIMM age, workload and vendor.

3.5 Q — quantum

IN PLAIN ENGLISH

This term is 1 by policy for workloads declared classical. For a workload that declares a nonzero non-Clifford budget, Q is a model-specific capability and price field. It is not a universal measure of quantum advantage.

Stabilizer circuits built from Clifford operations admit polynomial-time classical simulation; polynomial does not automatically mean inexpensive for every problem size [3]. **PUBLISHED** Holotrader's classical routing and accounting code does not require a quantum device. That software fact is separate from any claim that a future error-corrected quantum workload is easy.

The prototype assigns a declared non-Clifford count t the illustrative cost index 9^t . **MODELLED** Thus the index is 531,441 at $t = 6$ and about 1.2×10^{19} at $t = 20$. This is neither a lower bound for arbitrary circuits nor proof that classical exact or approximate simulation is impossible; cost depends on circuit structure, qubit count, error tolerance and algorithm.

Why this matters even if you never buy quantum capacity

It gives the prototype an explicit capability declaration. Classical machines run the exchange's ordinary routing and accounting layer. A future venue could quote separately for a verified quantum service, but no such substrate is provided here.

The prototype refuses to place a plan whose declared capability exceeds a node's declared capability. That is a policy invariant, not a physical impossibility theorem. The refusal code is implemented in software and in the Boolean admission primitive described in §7.

3.6 L — locality

L prices the prototype's address-distance from a declared data anchor. Inside one complete cell, graph distance is exact. Across recursive address levels it is a modeled metric, not measured

network latency. §4 separates the two.

3.7 The floor: what the exchange refuses to do

A two-sided pricing mechanism with an aggressive discount needs a floor, or the balancer will happily discount a machine to a price that never repays it.

$$\text{floor} = \text{electricity} + \text{maintenance reserve} + \text{capital recovery}$$

In the prototype, electricity is computed from modeled draw, interval price and PUE; maintenance reserve and capital recovery are policy functions. **MODELLED** Production use would need accounting treatment, residual value, utilization, service history and safeguards against double-counting the health discount.

IN PLAIN ENGLISH: What a maintenance reserve is

Money withheld from current revenue for expected future service. The prototype maps its modeled hazard to a reserve; it has not estimated a causal maintenance cost attributable to a particular buyer.

The prototype **will not post an ask below the floor**, and a regression test asserts that quote invariant. This is an ask-price policy, not evidence of a market-clearing equilibrium.

4 The network is the computer

This section separates an exact 40-vertex graph from the prototype’s proposed physical and recursive interpretations.

4.1 The identity

Holotrade’s cell addresses use $W(3,3)$, the symplectic generalized quadrangle over the four-dimensional vector space over $\mathbb{F}_3$. Its collinearity graph has 40 vertices and 240 undirected edges [9].

IN PLAIN ENGLISH: What that structure is, without the jargon

Start with the 80 nonzero four-digit vectors over $\mathbb{F}_3$. Identify a vector with its nonzero scalar multiple, leaving 40 projective points; the implementation chooses the representative whose first nonzero coordinate is 1. Two distinct points are adjacent when the alternating form below vanishes.

The exact graph properties used by the prototype are **PROVED** and exhaustively checked in the repository:

Property	Value	Why a buyer cares
Degree	12	Each vertex has twelve graph neighbours. A physical design would have to provision or emulate those links.
Diameter	2	Every pair is at graph distance at most two in the intact, unweighted graph. This says nothing about congestion, queueing or failed links.
Non-adjacent paths	4	Every non-adjacent pair has four distinct common neighbours, hence four internally vertex-disjoint two-edge paths.
Minimum bisection	100 of 240	Every $20 20$ cut crosses at least 100 edges, and an explicit cut attains 100. Link rate and traffic determine bandwidth.
Adjacency table	none	Adjacency is one arithmetic predicate. Choosing among relays, responding to failures and balancing traffic are separate routing tasks.

Address decides adjacency, not the whole route

For canonical point representatives u and v , the predicate is

$$\langle u, v \rangle = u_0v_1 - u_1v_0 + u_2v_3 - u_3v_2 \pmod{3}$$

and two distinct machines are adjacent exactly when it is zero. For a non-adjacent destination, software still selects one of four relays; a deployed network would also need failure,

congestion and policy handling.

4.2 The consequence: you buy shapes, not counts

Compute throughput and link bandwidth remain different physical resources, but a basket quote can expose their topology jointly. The prototype therefore sells a modeled *shape adjustment* in addition to a node count.

	40 scattered machines	40 machines as one cell
What it is	40 capacity positions with an arbitrary owned subgraph	The complete 40-vertex cell
Worst-case graph distance	may be disconnected	2
Minimum balanced cut	depends on the owned graph	exactly 100
Two-edge paths	depends on placement	4 for non-adjacent pairs

The same machine count can therefore expose different topology. Holotrade’s current *normalized induced-edge coherence* compares the number of graph edges induced by a basket with the model’s contiguous-basket baseline. It is a basket property, never a node property.

The current basket-price multiplier is superlinear by design. **MODELLED** A production calibration would require workload traces and measured link value.

Exact full-cell bisection; modeled partial-basket coherence

The spectral lower bound gives 100 for a balanced cut of the complete graph. In the repository’s canonical point order, the 20-point side

$\{2, 4, 5, 7, 8, 9, 11, 12, 16, 17, 19, 21, 23, 24, 28, 29, 32, 35, 38, 39\}$

has exactly 100 crossing edges, so the bound is attained. **PROVED** For partial baskets, the implementation uses normalized induced-edge coherence, not the minimum bisection of the induced graph. **MODELLED**

4.3 Which topologies are actually optimal, and can they be delivered

The bisection result above is about the whole cell. A scheduler does not reserve the whole cell — it reserves a subset — so the operational question is sharper: for a request of *m* nodes, what is the best possible shape, does one exist, can it be placed on whatever is free, and what happens to the leftovers.

All four have exact answers. **PROVED**

IN PLAIN ENGLISH: What "optimal shape" means here

Two reservations of the same size can differ enormously in how well connected they are internally. The extremes are worth naming: the *densest* shape of a given size packs in as many internal links as the geometry permits, which is what one tightly coupled job wants; the *most spread* shape is the one every outside node sees equally, which is what failure-independent replicas want. Everything else sits between them.

Writing $\mathbf{1}_T$ for the indicator of a candidate set and $f = \mathbf{1}_T - \frac{m}{40}\mathbf{1}$, the spectral bounds are attained exactly when f lies wholly in one eigenspace. Using the strongly regular identity $A^2 = 8I - 2A + 4J$, that turns into a pure counting condition on neighbourhoods, and the condition is decidable by exhaustive search:

		condition on every node				forces	
densest ($f \in E_2$)		$ N(v) \cap T = 2 + m/4$ inside, $m/4$ outside				$4 \mid m$	
most spread ($f \in E_{-4}$)		$ N(v) \cap T = 2m/5 - 4$ inside, $2m/5$ outside				$5 \mid m$	

m	$\max e(T)$	$b(T)$	# shapes	τ	busy	tolerated	tilings
4	6	36	40	11		10	36
8	16	64	585	8		7	57,132
12	30	84	5,400	8		7	—
16	48	96	21,330	6		5	—
20	70	100	33,264	6		5	16,632
24	96	96	21,330	4		3	—
28	126	84	5,400	4		3	—
32	160	64	585	3		2	—
36	198	36	40	2		1	—

Four things in that table are worth reading off.

A densest shape exists at every multiple of four, and nowhere else. So whenever a request is a multiple of four there is a provably optimal shape to serve it, and the repository holds a witness for each.

The shape counts are a palindrome — 40, 585, 5,400, 21,330, 33,264, 21,330, 5,400, 585, 40. That is forced rather than coincidental: the complement of a densest shape is a densest shape, so the count at m must equal the count at $40 - m$. The search reproduced it without being told, which is a real check on the search.

The four-node shapes are exactly the 40 lines of the geometry. Verified three ways, including brute force over all $\binom{40}{4} = 91,390$ subsets. The optimal small reservation is not something the design imposed; it is the incidence structure's own object.

Perfectly spread sets exist only at $m = 20$. Every other admissible size is impossible, not merely unfound.

4.3.1 Delivery: guarantees rather than averages

A catalogue nobody can place is a wall poster. The symmetry group makes shapes movable: an automorphism preserves adjacency, so it preserves internal link count and boundary *exactly*, and a shape transported onto free nodes carries its optimality with it as a matter of definition rather than approximation.

The column marked **busy tolerated** is the part a scheduler can act on. It is $\tau - 1$, where τ is the smallest number of blocked nodes that defeats *every* placement of the shape. Below it a placement is certain regardless of *which* nodes are busy; at it, an adversary can block the shape entirely. This is a worst case, not an average.

Why the four-node guarantee is exactly ten

The densest four-node shapes are the 40 lines, so defeating them means hitting every line. Each point lies on 4 lines, so k points cover at most $4k$ incidences, and covering all 40 needs $k \geq 10$. *Equality would require all 40 incidences to be distinct* — no two of the ten points collinear — which is precisely an **ovoid**.

This quadrangle has none: its independence number is 7, strictly below the Hoffman ratio bound of 10, established here by exhaustive search rather than by citation. Hence $\tau \geq 11$, and $\tau = 11$ is achieved.

So a four-node reservation **survives any ten simultaneous busy or failed nodes**, and eleven specific ones can defeat it. The bound is tight both ways, and an abstract non-existence result is what fixes an operational number.

The same non-existence caps failure-independent placement: **no eight nodes share zero links**, at any price, on any fabric of this shape. Seven is the ceiling on replica spreading.

4.3.2 Fragmentation: the sizes that leave nothing stranded

A tiling is the fragmentation-free schedule — every node used, every reservation provably densest, nothing left between jobs. It needs $4 \mid m$ so a densest shape exists, and $m \mid 40$ so the pieces exhaust the cell, which admits only $m \in \{4, 8, 20, 40\}$ before any search runs.

All three non-trivial cases tile, and the counts are exact: 36 at $m = 4$, 57,132 at $m = 8$, and 16,632 at $m = 20$. The first are the classical *spreads* of the quadrangle, ten disjoint lines meeting every point-star exactly once. The third is exactly half the number of 20-shapes, because each tiles with its own complement — the palindrome above, showing up again as a factor of two.

An admission policy, not an inequality

Reserve in units of 4, 8 or 20 and the cell partitions perfectly, with zero stranded capacity and every piece provably densest. At any other size some capacity is stranded by arithmetic: 4 nodes at $m = 12$, 8 at $m = 16$, and more above. **PROVED**

Scope. These are exact statements about the level-1 point graph: reservations of *points*, verified by three independent routes — combinatorial search in JavaScript, eigenspace projection in Python, and GRAPE/nauty in GAP, which also settles that the graph’s full automorphism group has order 51,840 and that every shape orbit is unchanged under it, so the guarantees are exact rather than conservative. They say nothing about link-state routing, congestion, failure recovery, or placement across an address hierarchy, none of which is built.

4.4 What happens above one cell, and why it does not scale

Everything above concerns a single 40-node cell. A fleet is not one cell, so the question is what survives when the fabric is built recursively. The answer is sharp, and it is mostly negative.

PROVED

IN PLAIN ENGLISH: The recursive fabric

Each of the 40 positions is replaced by a whole copy of the structure, giving 40^n leaves at level n . Two leaves are adjacent when they sit in the same cell and are adjacent inside it, or when they sit at the same position in adjacent cells. That is the homogeneous wiring — no gateway node, no privileged position.

At level n the fabric is $12n$ -regular with second eigenvalue $12n - 10$, so

$$k - \lambda_2 = 10 \quad \text{at every level}$$

The spectral gap never grows. Everything below follows from that one fact.

n	leaves	degree	λ_2	gap	expansion	smallest shape	survivable
1	40	12	2	10	0.8333	4	25.00000%
2	1,600	24	14	10	0.4167	160	0.62500%
3	64,000	36	26	10	0.2778	6,400	0.01562%
4	2,560,000	48	38	10	0.2083	256,000	0.00039%
5	102,400,000	60	50	10	0.1667	10,240,000	0.00001%

4.4.1 Every optimal shape above level one is a replication

A set attains the densest bound exactly when its indicator, minus a constant, lies wholly in the λ_2 -eigenspace — and for a Cartesian power that eigenspace consists of a level-1 eigenvector placed in *one* coordinate. So the indicator must be a sum of functions of separate coordinates. Since it takes only the values 0 and 1, and $|A_1 + \cdots + A_n| \geq \sum |A_i| - (n - 1)$ for finite sets of reals, at most one of those functions can be non-constant.

The classification, and what it forbids

Every densest reservation above level one is a level-1 optimal shape replicated along one coordinate. **There is no third kind**, so there is no cleverer shape waiting to be found.

Three consequences, all bad for large fleets. Expansion decays as $10/(12n)$, so the fabric gets structurally worse at connecting itself as it grows. The ladder of available sizes is $40\times$ coarser at each level — at level 5 the smallest provably densest reservation holds **10,240,000** leaves and there is nothing below it. And every one of them tolerates the same **ten** simultaneous failures, which is 0.00001% of a level-5 fabric.

The practical reading is that provably-densest reservations are a **level-1 tool**. Above that, the honest move is to compose level-1 shapes inside cells and accept a sub-optimal joint. The implementation says exactly that: `levelAdvice()` refuses off-ladder requests, reports the survivable fraction, and recommends the alternative rather than quoting a bound nobody can use.

Prior art. None of this machinery is new. Intriguing and tight sets are Payne (1987) and Bamberg–Kelly–Law–Penttilä; the same objects appear as equitable 2-partitions, where the eigenfunction characterisation used here is standard, and equitable 2-partitions of Cartesian powers have their own literature. The separability step is elementary and likely folklore. What is claimed is not new mathematics but that *this* fabric is now classified, so the architectural question has a definite answer.

Scope. These are statements about the level-1 point graph and its Cartesian powers. The repository’s software distance model, $16n - 14$, measures a different quantity — a declared cost in reversible control moves — and the measured graph diameter at level 2 is 4. Both are fine; conflating them is not.

4.5 The mechanism this creates: a defragmentation market

IN PLAIN ENGLISH

Trading entitlements can fragment a participant’s accessible induced subgraph, even though it does not change the physical links. The demo seeds fragmented portfolios to illustrate this allocation problem. **MODELLED**

The fix is a **swap book**. I give you my isolated machine sitting in *your* block; you give me yours sitting in *mine*. No cash changes hands beyond a small adjustment for the difference in condition between the two machines. The prototype offers a swap only when both modeled portfolio scores rise. Physical network capacity does not change.

This resembles entitlement defragmentation. Real acceptance would also depend on hardware class, data locality, legal constraints and transaction costs.

A swap is only offered when both sides gain

The engine computes both parties’ normalized induced-edge score before proposing a swap, and a regression test asserts both scores increase. This is an invariant of the prototype objective, not proof of economic welfare.

4.6 The recursive address model

IN PLAIN ENGLISH

A level- n software address is an n -digit base-40 path. The current distance model charges at most two graph edges at the first divergent digit and eight abstract reversible moves for every remaining suffix digit on both sides. It does not yet specify a physical graph product, switches or links.

The address count 40^n is exact. Under the current software metric, equal-depth addresses have the bound

$$d_n \leq 2 + 16(n - 1) = 16n - 14.$$

MODELLED

This is an address-cost bound, not a physical hop theorem.

Level	Machines	Worst-case hops	Seats
1	40	2	one address cell
2	1,600	18	modeled
3	64,000	34	modeled
4	2,560,000	50	modeled
5	102,400,000	66	modeled
6	4,096,000,000	82	modeled
7	163,840,000,000	98	modeled

One prototype interface at every modeled scale

A software **Composite** exposes the same quote interface as a leaf: an address, modeled history, health, utilisation and throughput. The same pricing function can therefore quote both without type-specific branches.

The repository test quotes a leaf, cell and composite through that one code path. This proves interface closure only; it does not prove that a physical campus is one machine or that liquidity aggregates self-similarly.

Any deployed service-level agreement would still need measured latency, availability, congestion and remedies. The model's address bound is not an SLA.

4.7 Joining is local in the prototype

A machine can fill an unused modeled address slot with a local registry update. A complete $W(3,3)$ cell has exactly 40 positions, so a 41st machine requires another cell, a replacement or a different topology. Physical joining would also require cabling, link configuration, identity and failure handling.

Point transitivity means graph positions are structurally equivalent; it does not make heteroge-

neous machines operationally indistinguishable. The local splice test is therefore an implementation invariant, not a zero-cost claim.

5 Instruments

The prototype renders five order or contract templates. Only the first is a spot capacity purchase; the supply offer is seller-side. Forward, option and lease prices below are scenarios, not validated valuation models. They would require delivery, assignment, collateral, cancellation, exercise, default and jurisdiction-specific regulatory terms.

Instrument	Tenor	The risk it addresses
Spot second	node- immediate	Capacity metered and settled by the second, with an hourly-equivalent display rate and a separate cold-start charge.
Forward block	1–90 days	An illustrative reservation for future capacity. Compute is not a freely storable commodity, so the displayed spot-plus-risk rule is a model rather than a no-arbitrage cost-of-carry theorem.
Burst option	1–30 days	You <i>might</i> need 400 machines on launch day. You want the right, not the obligation. A production premium requires a specified payoff distribution, exercise rule, reservation policy and capacity correlation; the demo uses a simple scenario rule.
Genome lease	7–365 days	You want versioned software state with a history on your workload class. Its premium remains a hypothesis pending out-of-sample workload validation.
Supply offer	open	You own idle machines and post capacity subject to the modeled quote floor. This is an order type, not a buyer-risk instrument.

The margin hypothesis

Competitive undifferentiated capacity may face pressure toward marginal and capital-recovery cost, but convergence is not guaranteed and the prototype is not evidence of market structure.

Reusable software state could support differentiation if it produces verified, portable, out-of-sample improvement after accounting for contamination and customer-data restrictions. That is the main commercial hypothesis to test, not a result established here.

5.1 A policy mobility score, not observed liquidity

IN PLAIN ENGLISH

Market liquidity depends on counterparties, order flow, prices, substitutability and time. Before trading starts, the rule engine can compute only a narrower quantity: the fraction of address positions permitted by declared policy.

Let N_{eligible} be the number of the 40 cell positions allowed by a specific prototype policy configuration. The displayed mobility score is $M = N_{\text{eligible}}/40$. **MODELLED** The eligible counts below are policy scenarios. They are not group orbits, stabilizer quotients or measurements of market liquidity.

Policy scenario	Eligible	M	Display band
none	40	1.000	deep
GPU affinity	27	0.675	liquid
single-tenant	20	0.500	liquid
data residency	13	0.325	thin
air-gapped	5	0.125	bilateral

What the score is useful for

M is an auditable policy disclosure: it tells a buyer how many modeled positions satisfy the declared constraints. It can flag a potentially narrow delivery set. It must not be labelled liquidity or used as a price discount until calibrated against actual fills, spreads and market depth.

6 Energy and the thermodynamic floor

IN PLAIN ENGLISH

Compute is bought in dollars and paid for in joules. Data centres drew about 415 terawatt-hours globally in 2024 — roughly 1.5% of world electricity. The International Energy Agency projects about 945 TWh by 2030 under its base case [2]. PUBLISHED

That is a scenario, not a physical law. Holotrader's narrower claim is that a quote can expose the energy price interval, PUE assumption and modeled joules instead of hiding them in one rate.

6.1 The honest efficiency denominator

IN PLAIN ENGLISH: The Landauer limit

Landauer derived a lower bound $kT \ln 2$ for erasing one bit under the stated thermodynamic assumptions [5]. Bennett showed how logically reversible computation can in principle avoid compulsory erasure at each logic step [6]. Neither result removes loss, control, speed, noise or error-correction costs from a physical device.

The source model assigns 58 ternary records per error-correction cycle and assumes each record is eventually reset, with conditional lower bound $kT \ln 3$ per reset. Measurement alone is not necessarily erasure; record reset is the relevant step. Under that model, at room temperature the arithmetic is

$$2.64 \times 10^{-19} \text{ joules per modeled cycle at 300 K}$$

MODELLED

This per-cycle conditional bound cannot be compared directly with a vendor's joules per operation until "cycle", useful operation, clock rate and all control and memory costs are mapped consistently.

How the denominator may be used

The Landauer expression is a stable physical reference once the logical erasure event is defined. It is not a workload-efficiency benchmark and can mislead if vendors count different operations or omit memory, cooling and control.

The demo displays a conditional orders-of-magnitude gap for exploration. A production quote should pair it with useful-work-per-joule, total facility energy and the exact operation definition.

Honest scope. The number is conditional on the model's 58 reset records. It is not measured device efficiency, a throughput result or evidence that only those erasures occur. No full build of the proposed architecture exists.

7 The hardware

The prototype’s combinational routing and admission-policy classifier is small enough to synthesize for an FPGA. It consumes predicate results; it does not perform digital-signature verification, artifact hashing, replay-store lookup or remote attestation itself.

The RTL computes the cell adjacency/migration fields and applies a fixed priority to supplied admission predicates. Both are implemented in `rtl/holotrade_admit.v`.

7.1 It is proved, not just written

IN PLAIN ENGLISH: What formal verification is

Testing evaluates selected executions. Here, *formal equivalence* means a SAT solver found no input on which an encoded RTL circuit differs from its encoded reference. That is exhaustive for the stated finite input model; it is not proof that upstream predicates are truthful or that the reference captures all deployment requirements.

Two independently written implementations of the form — staged $\mathbb{F}_3$ operators and integer arithmetic with one final reduction — are equivalent over all $2^{16} = 65,536$ encoded address pairs. A second miter checks the full combinational admission and refusal-priority function over all $2^{25} = 33,554,432$ encoded inputs, including invalid-address handling and the seventh refusal code `BAD_ADDR`. **PROVED**

SAT proof finished - no model found: SUCCESS!

Equivalence is necessary and *not sufficient*: a reference can encode the wrong requirement. The software suite separately maps the 40 canonical points to the RTL representation and exhaustively checks all 1,600 ordered point pairs. Security predicate generation remains outside this proof boundary.

7.2 Current synthesis footprint

Synthesised to an iCE40 HX8K field-programmable gate array:

$49 \times \text{SB_LUT4}$ $6 \times \text{SB_CARRY}$

MEASURED

This footprint covers address validation, form/adjacency, migration cost and the Boolean refusal-priority classifier. It excludes the cryptographic and stateful systems that produce `sig_ok`, `nonce_fresh` and `pins_ok`.

What is deliberately not reported

This is Yosys iCE40 synthesis only. No place-and-route timing, clock frequency, power, latency or hardware measurement is reported; those are properties of a placed-and-routed design on a specific part.

7.3 The gate refuses rather than degrades

Seven refusal codes are included in the fixed priority function. The RTL classifies already-computed predicates; the full 2^{25} miter checks the code and priority for every encoded input.

#	Refusal	Meaning
1	BAD_SIG	The upstream signature-verification predicate is false.
2	REPLAY	The upstream replay-store predicate reports a used nonce.
3	WINDOW	The supplied time-window predicate is false.
4	PIN_DRIFT	The supplied artifact-integrity predicate is false.
5	NO_MAGIC	Declared workload capability exceeds declared node capability.
6	IN_SERVICE	The node is unavailable for placement.
7	BAD_ADDR	One or both encoded addresses are not canonical valid points.

There is no best-effort output in this classifier: it admits or emits one code. Deployment assurance still depends on authentic predicate sources, a trusted boot path and preventing bypass or FPGA reconfiguration.

8 What exists today

Component	Status	Evidence
Cell geometry & adjacency	PROVED	exhaustive 40-point graph checks; SAT form equivalence
Six-term pricing engine	MEASURED	implemented decomposition and floor invariants
Two-sided balancer	MODELLED	paired 64-seed result reported with confidence intervals
Coherence & swap book	MODELLED	induced-edge objective; both modeled scores must rise
Recursive address metric	MODELLED	40^n count; software bound $16n - 14$; no physical product graph
Plan seal & admission policy	MEASURED	demo integrity seal; seven predicate-input refusal codes
Per-second metering	MEASURED	prototype prorates hourly-equivalent quotes by node-second
Contract templates	MODELLED	scenario pricing; no live market or legal contract
Trading terminal	MEASURED	browser prototype with seed data
RTL policy primitive	PROVED	form and full admission miters; synthesis 49 LUT4 + 6 carry
Physical photonic substrate	—	does not exist ; not required by anything above
Real market data feeds	—	seed data; engines are feed-agnostic by construction

8.1 The reproducibility contract

The repository checks the finite and implementation claims with:

```

npm test                finite checks and prototype contracts
npm run verify:rtl      SAT-prove form and admission miters
node experiments/balancer_ab.js -summary reproduce the paired 64-seed
                        packet

```

The simulation uses a seeded pseudo-random generator. The frozen headline packet in `data/balancer_ab_64.json` uses 64 paired seeds, 220 nodes and 500 one-minute steps (8.333 simulated hours per arm); the same seed initializes each on/off pair.

Two kinds of test, deliberately not mixed

Substrate facts are exact finite mathematics with single correct answers. Any deviation is a bug, not a tuning question.

Engine contracts are implementation invariants: the floor is not breached, capability refusal has a reason, middle-entry deletion breaks the demo chain, and one quote interface accepts leaves and composites.

Model experiments depend on coefficients and are reported separately. A test that merely returns or recomputes a chosen constant establishes code consistency, not an external theorem or empirical result.

8.2 Two bugs the tests caught, and why that is the point

Both were found by the suite during development and both are the kind that survive casual review.

A doubled topology score. An earlier edge normalization used the wrong denominator. The corrected full-cell normalization is 100, while partial-basket outputs are now described accurately as induced-edge coherence rather than actual bisection.

A non-monotone price curve. The demand multiplier had a discontinuity at one of the seams between its discount, band and premium regions, creating a narrow range of utilisation where a machine got *cheaper by becoming busier*. That is directly arbitrageable: load the machine you are about to rent, then rent it at a discount. A monotonicity test caught it, and the fix is documented in the source beside the code it corrects.

9 Honest scope

This section exists because a document that only argues one direction is not worth the paper. Every claim here is sorted by what actually supports it.

Exact and computed in this repository

The $W(3, 3)$ geometry — 40 points, 240 edges, 40 lines of 4 points each, diameter 2, bisection exactly 100 as the spectral bound met by the displayed explicit cut, and symmetry-group order 51,840. The 40^n recursive address count. The form miter over 2^{16} encoded pairs, the full admission miter over 2^{25} encoded inputs, and the 1,600-pair canonical-point cross-check. These are statements about finite software objects and encodings. They do not establish physical topology, timing, security of upstream predicates or market performance.

Modelled, and never presented as measured

The fleet, wholesale prices, accumulated genomes, wear trajectories, order flow, partial-basket coherence, recursive distance, 9^t cost index, 58-reset Landauer calculation and contract valuations are models. This is a seeded simulation, not a production system with customers.

The modules expose feed interfaces. Real integration would still require units, timestamps, missing-data policy, authentication, settlement reconciliation, calibration and changes discovered during deployment.

Asserted by the source programme, not by this product. The $W(3, 3)$ - E_8 substrate programme makes claims about $W(3, 3)$ being a candidate physical substrate for fundamental physics. **Holotrader does not depend on any of that being true.**

The cell graph and its adjacency rule are finite mathematics independent of the physics programme. Error-correction interpretation, thermodynamic accounting, recursive physical topology and quantum capability pricing add assumptions and are labeled as models here.

The photonic hardware that would realise the quantum layer does not exist. What runs here is ordinary classical software for the exchange and finite graph. The quantum field is a declared capability and illustrative price index, not a capability being offered.

9.1 The risks worth stating

- **Supply-side cold start.** A two-sided market needs machines and buyers. The prototype neither establishes safe lending nor makes physical joining free, and it does not create either side of the market.
- **The genome premium must be demonstrated, not argued.** The claim that a specialised core finishes in measurably fewer hours is plausible, grounded in existing practice, and *not yet measured on real workloads*. It is the single most important thing to validate, because it is where the durable margin lives. Until it is measured on production jobs, it is a hypothesis with a good mechanism.

-
- **Per-second settlement is an operational commitment.** Metering every running instance every second against a live grid price is real infrastructure with real failure modes, and the reconciliation problem when a meter disagrees with an invoice is not solved by having the right formula.
 - **Regulatory surface.** Transferable forwards and options can create financial, commodities, securities, licensing, tax and consumer-protection issues. Product classification requires qualified counsel in every offered jurisdiction; this paper provides no legal conclusion.

A Glossary

Term	Meaning
Adjacency	Two graph vertices sharing an edge. Here it is determined by the symplectic predicate on canonical addresses.
Bisection	The minimum number of edges crossing any balanced partition of a graph. It is a topological quantity; physical bandwidth also needs link rates.
Clifford operations	Operations in stabilizer circuits that admit polynomial-time classical simulation under the Gottesman–Knill framework. Polynomial does not mean free.
Coherence	The prototype’s normalized induced-edge basket score. It is not actual partial-graph bisection or measured bandwidth.
Content-addressed	Identified by a digest of content. Substitution is detectable only if a trusted verifier recomputes and authenticates the expected digest.
Correctable error	An error within the correction capability of the relevant ECC scheme. Counts are candidate reliability features, not deterministic failure predictions.
Derate	The gap between a machine’s nameplate performance and what it actually delivers today, after wear.
Diameter	The worst-case number of hops between any two machines in a network.
Firecracker	Amazon’s open-source microVM hypervisor, which runs their serverless products.
Gini coefficient	A number from 0 to 1 measuring how unevenly something is distributed. Used here for load across a fleet; lower is better.
Hash chain	A log whose entries commit to their contents and predecessor. Verification must recompute hashes and trust or externally anchor a chain head.
Hypervisor	Software that creates and isolates virtual machines on physical hardware.
Landauer limit	The thermodynamic minimum energy to erase one bit, about $kT \ln 2$. A physical law, not an engineering target.
MicroVM	A virtual machine with a reduced device model and separate guest kernel. Isolation still depends on implementation and deployment.
Mobility score	The modeled fraction of cell positions eligible under declared policy. It is not group orbit size or observed liquidity.
Node-second	One machine allocated for one second, charged at one 3,600th of the displayed hourly-equivalent rate plus itemised fees.
Non-Clifford / magic	Operations outside the Clifford stabilizer set. Their classical simulation cost depends on the circuit and algorithm; 9^t is this prototype’s index.
Pin drift	An observed artifact no longer matching its expected digest. Production enforcement requires trusted resolution, hashing and attestation.

Term	Meaning
PUE	Power Usage Effectiveness: total facility power divided by computing power. A straight multiplier on the electricity bill.
Splice	Filling an unused modeled address slot with a local registry update. Physical integration is out of scope.
Symplectic form	The arithmetic expression on two canonical point representatives whose vanishing defines graph adjacency. It does not select a full route.
Thermal cycling	Repeated heating and cooling, one of several mechanisms that can contribute to hardware wear.
Weibull hazard	A parametric lifetime model. A fitted shape above 1 represents increasing hazard; the prototype coefficient is not field-calibrated.

B Selected constants and parameters

The graph constants below are exact for $W(3, 3)$. The remaining rows are explicit inputs inherited from the source programme or chosen by the prototype; they are not forced engineering quantities.

Value	Symbol	Status	Role
3	q	exact	field order / address radix
2	λ	exact	common neighbours of an adjacent pair
4	μ	exact	common neighbours of a non-adjacent pair
12	k	exact	graph degree
40	v	exact	vertices per complete cell
240	$ E $	exact	undirected edges per complete cell
100	—	exact	minimum balanced edge cut of the complete cell
51,840	$ \text{Aut} $	exact	full graph-symmetry order used by the source model
27/80	—	source-model input	declared code-rate parameter; not a measured venue throughput cap
1/10	—	source-model input	contextual fraction; not production attestation evidence
58	—	conditional model	reset-record count used in the Landauer calculation

C References

References

- [1] A. Agache, M. Brooker, A. Iordache, A. Liguori, R. Neugebauer, P. Piwonka, and D.-M. Popa, “Firecracker: Lightweight Virtualization for Serverless Applications,” *17th USENIX Symposium on Networked Systems Design and Implementation*, 2020, pp. 419–434.
- [2] International Energy Agency, *Energy and AI*, Paris, 2025.
- [3] S. Aaronson and D. Gottesman, “Improved Simulation of Stabilizer Circuits,” *Physical Review A*, vol. 70, 052328, 2004; [arXiv:quant-ph/0406196](https://arxiv.org/abs/quant-ph/0406196).
- [4] B. Schroeder, E. Pinheiro, and W.-D. Weber, “DRAM Errors in the Wild: A Large-Scale Field Study,” *ACM SIGMETRICS/Performance*, 2009, pp. 193–204.
- [5] R. Landauer, “Irreversibility and Heat Generation in the Computing Process,” *IBM Journal of Research and Development*, vol. 5, no. 3, pp. 183–191, 1961.
- [6] C. H. Bennett, “Logical Reversibility of Computation,” *IBM Journal of Research and Development*, vol. 17, no. 6, pp. 525–532, 1973.
- [7] National Institute of Standards and Technology, *Digital Signature Standard (DSS)*, FIPS PUB 186-5, 2023.
- [8] Electric Reliability Council of Texas, “Real-Time Market,” official market information, ercot.com/mktinfo/rtm, accessed August 2026.
- [9] S. E. Payne and J. A. Thas, *Finite Generalized Quadrangles*, 2nd ed., European Mathematical Society, 2009.